4.2.19

Taraka Abhinav - EE25BTECH11016

Question:
Find the direction and normal vectors of each of the following lines

$$y = -x + 2 \tag{0.1}$$

Solution:

The given line equation can be rewritten in the standard form $ax + by = c$:

$$x + y = 2 \tag{0.2}$$

This equation can be expressed in vector form.
Let

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \tag{0.3}$$

The line equation can be written as:

$$\mathbf{n}^{\mathbf{T}}\mathbf{x} = c \tag{0.4}$$

where $\mathbf{n}$ is the normal vector to the line.

By comparing $x + y = 2$ with the vector form, we can identify:

$$\mathbf{n}^{\mathbf{T}} = \begin{pmatrix} 1 & 1 \end{pmatrix} \tag{0.5}$$

$$c = 2 \tag{0.6}$$

From $\mathbf{n}^{\mathbf{T}}$, we find that the **normal vector** is:

$$\mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{0.7}$$

The **direction vector**, $\mathbf{m}$, of the line is orthogonal to the normal vector $\mathbf{n}$. This means their dot product is zero, i.e., $\mathbf{n}^{\mathbf{T}}\mathbf{m} = 0$.

For a normal vector $\mathbf{n} = \begin{pmatrix} a \\ b \end{pmatrix}$, a direction vector can be $\mathbf{m} = \begin{pmatrix} b \\ -a \end{pmatrix}$. So, for our normal vector $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, the direction vector is:

$$\mathbf{m} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \tag{0.8}$$

We can verify this with the dot product:

$$\mathbf{n}^{\mathbf{T}}\mathbf{m} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = (1)(1) + (1)(-1) = 0 \tag{0.9}$$

The normal vector of the line is $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

The direction vector of the line is $\mathbf{m} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.